

Tirth Laheri

Data Scientist — Machine Learning Engineer — Data Engineer
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EDUCATION

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| Middle Tennessee State University
<i>Master of Science in Data Science GPA: 4.00/4.00</i>
• Relevant Coursework: Machine Learning, Statistical Methods, Data Mining, Data Visualization | Murfreesboro, TN
2024 – Present |
| Parul University
<i>Bachelor of Technology in Computer Science (IoT) GPA: 9.06/10.00</i>
• Relevant Coursework: Data Structures, Algorithms, Database Systems, Cloud Computing | India
2020 – 2024 |

EXPERIENCE

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| Graduate Assistant (Administrative)
<i>Physician Assistant Department, Middle Tennessee State University</i>
• Built a SQL-driven data pipeline to extract, transform, and automate real-time pass-rate and progression metrics for cohorts of 15–20 students , cutting reporting time by 90% .
• Wrote SQL queries and Excel-based automation to aggregate attendance, pass rate, and progression data across 3+ cohorts , eliminating manual calculation errors.
• Designed ETL and data validation workflows to streamline record intake and integrity checks between the database and reporting layer, reducing administrative processing time by 40% . | Aug 2025 – Present
Murfreesboro, TN |
| Data Science Intern
<i>SlashMark</i>
• Compared 4 classification models in Python and Scikit-learn on a 10,000+ record dataset, improving F1-score by 12% over the baseline model.
• Conducted exploratory data analysis and feature engineering across 20+ variables , identifying 5 key predictors that drove the majority of model performance gains.
• Delivered weekly analytical reports and visualizations in Matplotlib and Seaborn to a 4-person data science team, informing iteration on model selection and feature strategy. | Dec 2023 – Mar 2024
Hyderabad, IN |

PROJECTS

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| Multilingual Legal Document Intelligence <i>LangChain, OpenAI, pgvector, FastAPI</i>
• Develop a multilingual legal intelligence platform supporting 5+ languages and semantic search across 100+ legal PDF documents using OpenAI GPT and text-embedding-3-small .
• Built a RAG architecture with LangChain and PostgreSQL pgvector embeddings, achieving document retrieval latency of 50–500ms .
• Applied PaddleOCR with layout-aware parsing to extract structured information from multilingual legal documents, achieving over 95% OCR accuracy .
• Exposed scalable FastAPI services for OCR, semantic search, contextual Q&A, and document summarization, enabling real-time legal document analysis. | Jan 2026 – Present |
| Advanced Agentic RAG System <i>Python, LangChain, Multi-Agent, RAG, LLM, Streamlit</i>
• Developed a stateful 10-agent RAG workflow (Router, Retriever, Grader, Reformulator, WebSearcher, Synthesizer, Generator, FactChecker, SafetyChecker, Clarifier) with conditional routing to dynamically select retrieval, web search, or clarification per query.
• Implemented a self-correcting retrieval loop where a Grader agent scores document relevance and automatically triggers query reformulation or live web search via the Tavily API on low-confidence retrievals.
• Implemented post-generation FactChecker and SafetyChecker agents to verify claims via web search and filter unsafe content, and engineered a ChromaDB ingestion pipeline (PDF/DOCX/TXT/URL) behind a configurable Streamlit interface. | Mar 2024 |
| Diabetic Retinopathy Screening <i>Python, TensorFlow, Image Processing, Deep Learning</i>
• The Diabetic Retinopathy Screening system utilizes advanced neural networks, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) , for accurate image classification and early detection.
• Achieved high diagnostic accuracy (94.73%) through sophisticated image preprocessing, feature extraction, and deep learning techniques, ensuring sensitivity and specificity in detecting diabetic retinopathy. | Feb 2024 |

TECHNICAL SKILLS

Languages: Python, Java, SQL, C/C++, R, JavaScript, HTML/CSS

Machine Learning & Data Science: Machine Learning, Deep Learning, Statistical Analysis, Computer Vision, NLP, Exploratory Data Analysis (EDA), Feature Engineering

Generative AI & LLMs: LangChain, OpenAI API, Groq, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases (ChromaDB, pgvector)

Frameworks & Libraries: TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, FastAPI, Streamlit

Developer Tools & Platforms: Git, VS Code, Google Colaboratory, SQL, PostgreSQL, Power BI, Tableau, Cloud Computing

CERTIFICATIONS

• **Python for Machine Learning & Data Science** - Udemy Certification • **Machine Learning Specialization** - Coursera Certification • **Machine Learning Fast-Track Workshop** - Google Certification • **Cloud Computing** - Parul University • **Introduction to Internet of Things** - NPTEL Certification • **Web Development** - 1Stop.ai

PUBLICATIONS

Visible Light Communication Based Indoor Positioning System 2024

IEEE Xplore Digital Library

- Collaborated with a **4-member research team** to design and develop a VLC-based indoor localization system using photodiode sensors and microcontrollers for accurate indoor navigation.
- Improved positioning precision through light-intensity calibration and ML-driven error correction, validated through joint testing and peer review with co-authors.
- *Published in:* IEEE Xplore — ieeexplore.ieee.org/document/10601381

IoT-Based Remote Health Monitoring for Rural Clinics 2023

Private Research Repository, Parul University

- Designed an IoT-based patient monitoring system to collect real-time vitals (e.g., SpO₂, heart rate) for rural clinics.
- MQTT for secure data transmission and implemented rule-based anomaly detection to aid remote doctors.

Integration of Smart Wearables and Cloud Computing for Preventive Healthcare 2022

Private Academic Repository, Parul University

- Built a wearable-cloud architecture to track physiological signals and detect early risk indicators of chronic diseases.
- Implemented AI models to trigger health alerts, emphasizing preventive care and early intervention.